

Lab Manual

Digital Integrated Circuit Design

[ET3288]

Third Year Bachelor of Technology
Semester – V

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JULY 2026

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This is to certify that **Sudarshan Anil Chaudhari** of class **TY E&TC**, Division **G** Roll No. **06** have performed the above listed experiments in the **Digital IC Design** course Laboratory in the Department of **Electronics and Telecommunication Engineering** at Vishwakarma Institute of Technology, Pune.

Date:

In-Charge Faculty

Head of Department

Name: Sudarshan Anil Chaudhari

Roll No.: 06

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Division: G

EXPERIMENT NO: 01

TITLE OF EXPERIMENT: SPICE modeling of NMOS and PMOS transistor to plot I_{DS} Vs. V_{DS} and I_{DS} Vs. V_{GS} characteristics.

1.1 Aim: - Do SPICE modeling of NMOS and PMOS transistor to plot I_{DS} Vs. V_{DS} and I_{DS} Vs. V_{GS} characteristics.

1.2 Software/Hardware: - Any SPICE tool like Winspice, NGspice etc.

1.3 SPICE codes: (Copy paste your SPICE code here. Write your name and PRN No. in comments.) (Use $L = 90$ nm and $W = 1$ μ m for both transistors. Use $V_{DD} = 1$ V, $V_{DS,max} = V_{SD,max} = 1$ V, $V_{GS,max} = V_{SG,max} = 1$ V)

A) SPICE code for I_{DS} Vs. V_{DS} Characteristic of NMOS:

```
* NMOS IDS vs VDS
.model NMOS NMOS (LEVEL=1 VTO=0.4 KP=200u)
VGS G 0 0.5
VDS D 0 0
M1 D G 0 0 NMOS L=90n W=1u
.dc VDS 0 1 0.01
.control
run
plot -i(VDS)
.endc
.end
```

B) SPICE code for I_{DS} Vs. V_{GS} Characteristic of NMOS:

```
* NMOS IDS vs VGS
.model NMOS NMOS (LEVEL=1 VTO=0.4 KP=200u)
VGS G 0 0
VDS D 0 1
M1 D G 0 0 NMOS L=90n W=1u
.dc VGS 0 1 0.01
.control
run
plot -i(VDS)
.endc
.end
```

C) SPICE code for I_{DS} Vs. V_{DS} Characteristic of PMOS:

```
* PMOS ISD vs VSD
.model PMOS PMOS (LEVEL=1 VTO=-0.4 KP=200u)
* VSG = 0.5 V
VSG S G 0.5
* VSD = 0 to 1 V
VSD S D 0

* PMOS: Drain Gate Source Bulk
M1 D G S S PMOS L=90n W=1u
.dc VSD 0.001 1 0.01
```

```
.control
run
plot i(VSD)
.endc
.end
```

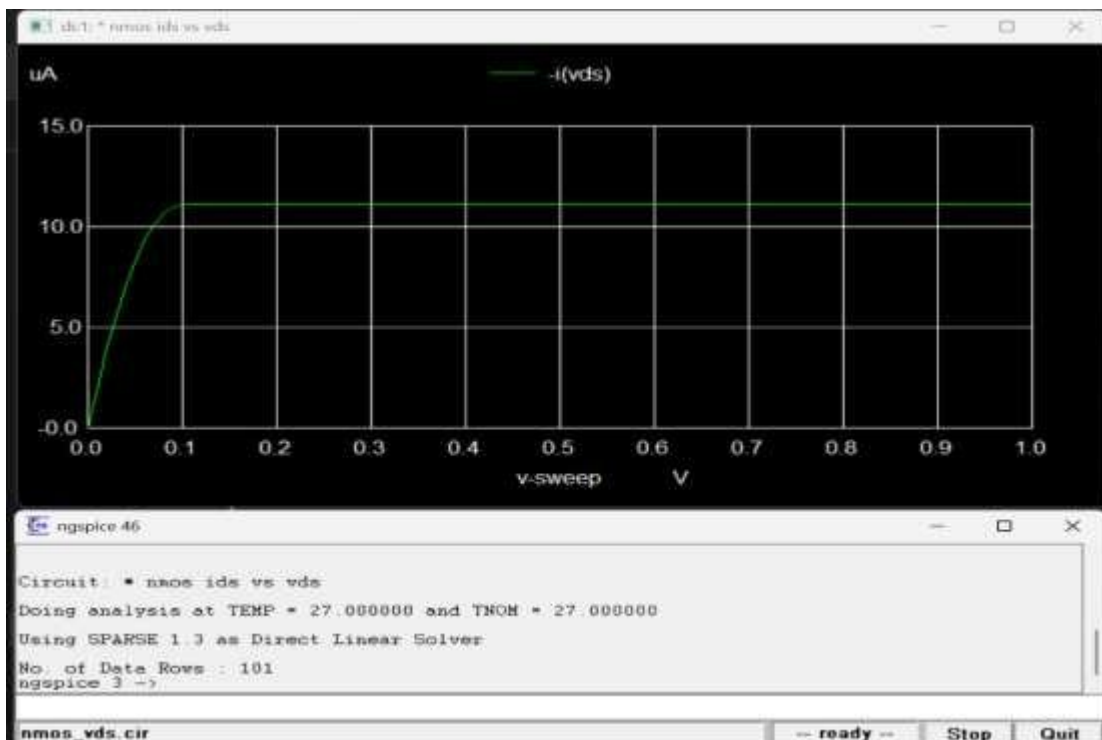
D) SPICE code for I_{DS} Vs. V_{GS} Characteristic of PMOS:

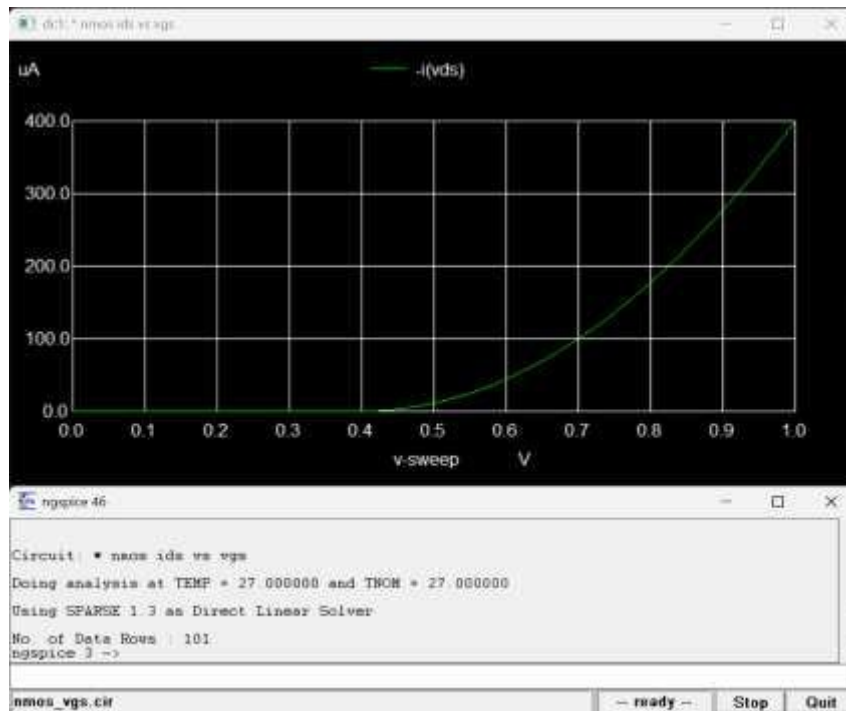
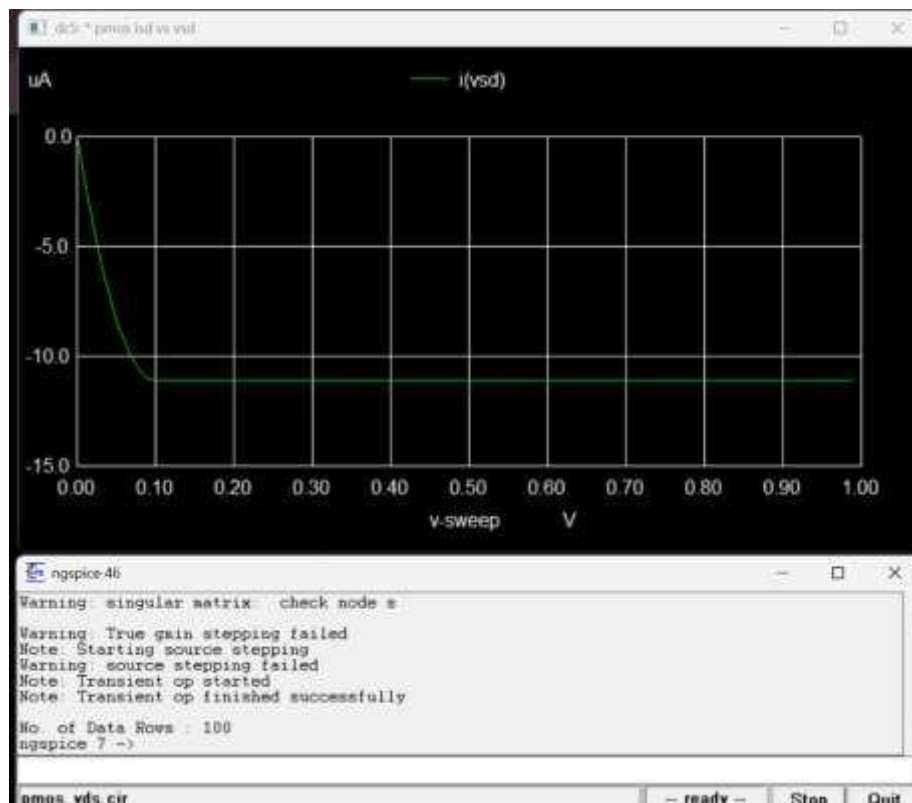
```
* PMOS  $I_{SD}$  vs  $V_{SG}$ 
.model PMOS PMOS (LEVEL=1 VTO=-0.4 KP=200u)
* Source = 1 V
VSS S 0 1
*  $V_{SG}$  swept from 0 to 1 V
VSG S G 0
*  $V_{SD} = 1$  V
VSD S D 1
* PMOS: Drain Gate Source Bulk
M1 D G S S PMOS L=90n W=1u
.dc VSG 0 1 0.01
.control
run
plot -i(VSD)
.endc
.end
```

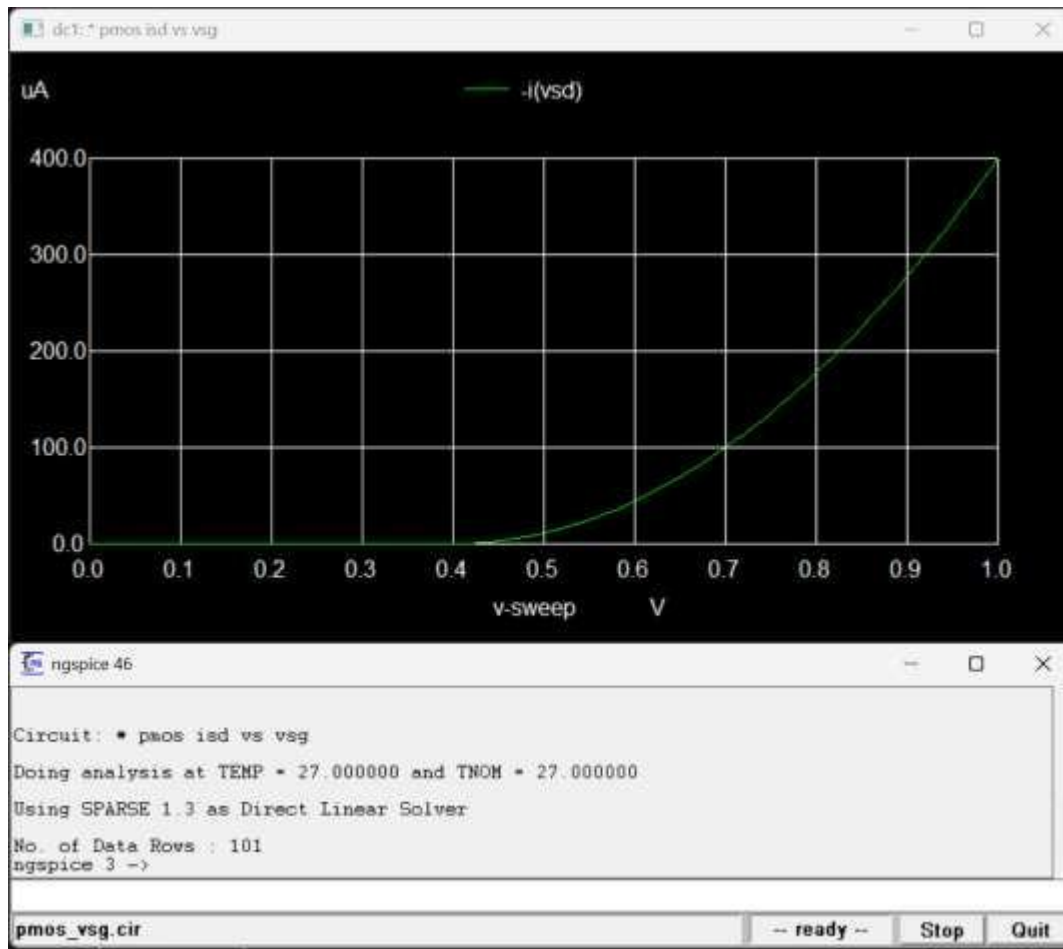
1.4 Simulation result screenshots: (Copy paste simulation result screenshots here.)

A) I_{DS} Vs. V_{DS} Characteristic of NMOS for $V_{GS} = 0.5V$:

$I_{DS, \max} = 11 \mu A$.



B) I_{DS} Vs. V_{GS} Characteristics of NMOS for $V_{DS} = 1V$:**Approx. $V_{th} = 0.4V$** **C) I_{SD} Vs. V_{SD} Characteristics of PMOS for $V_{SG} = 0.5V$:** **$I_{SD, max} = -11 \mu A$.**

D) I_{SD} Vs. V_{SG} Characteristics of PMOS for $V_{SD} = 1V$:**Approx. $V_{th} = 0.4 V$** **1.5 Conclusion:** Comparative comments on your results from both transistors

The SPICE simulation successfully verified the characteristics of NMOS and PMOS. Both show linear and saturation regions, with approximately 0.4 V threshold voltage. NMOS and PMOS have similar characteristics but opposite current directions.

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EXPERIMENT NO: 02

TITLE OF EXPERIMENT: Write SPICE model for CMOS inverter and perform transient and DC analysis.

2.1 Aim: - Write SPICE model for CMOS inverter and perform transient and DC analysis.

2.2 Software/Hardware: - Any SPICE tool like Winspice, NGspice etc.

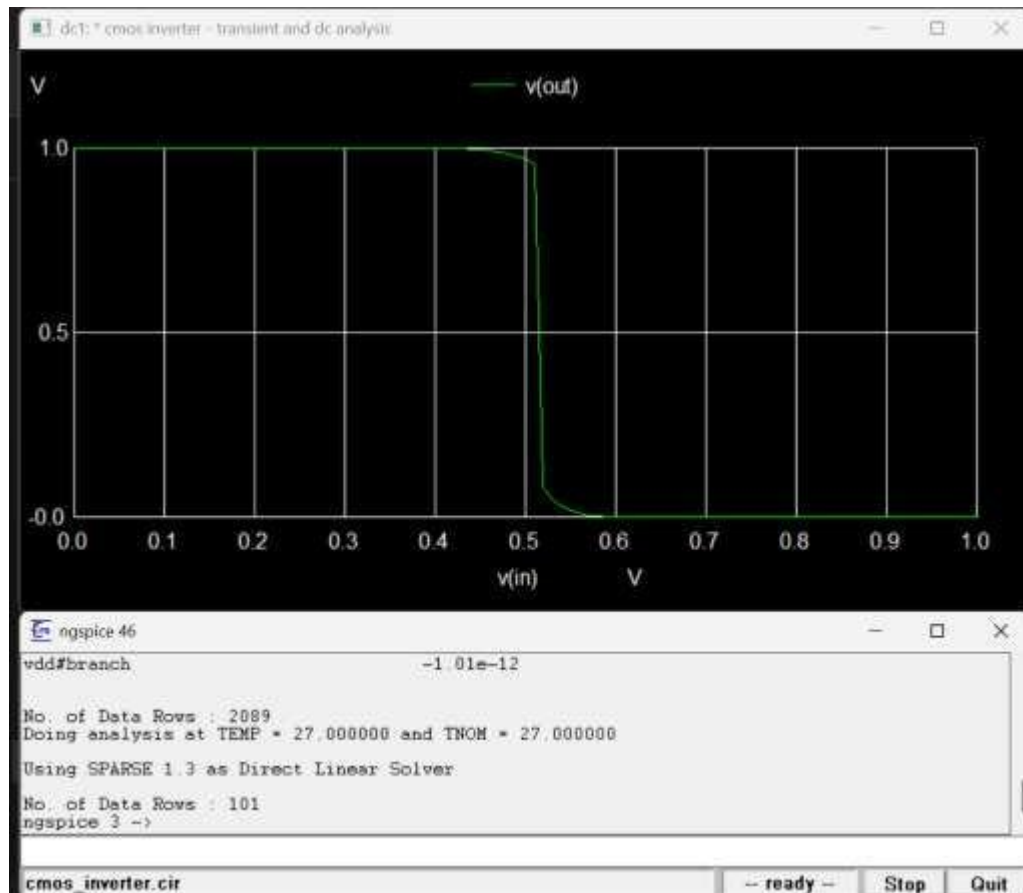
2.3 SPICE code: (Copy paste your SPICE code here. Write your name and PRN No. in comments.) (Use $L = 90\text{ nm}$ for both transistors, $W = 0.5\text{ }\mu\text{m}$ for NMOS and $W = 1\text{ }\mu\text{m}$ for PMOS. Use $V_{DD} = 1\text{ V}$)

```
* CMOS Inverter - Transient and DC Analysis
* Power supply
VDD vdd 0 DC 1
* Input signal for transient analysis
VIN in 0 PULSE(0 1 0 1n 1n 5m 10m)
* CMOS inverter
MP out in vdd vdd PMOS L=90n W=1u
MN out in 0 0 NMOS L=90n W=0.5u
* MOSFET models
.model NMOS NMOS (LEVEL=1 VTO=0.4 KP=200u)
.model PMOS PMOS (LEVEL=1 VTO=-0.4 KP=200u)
* Transient analysis
.tran 0.01m 20m
.control
* Transient analysis
run
plot v(in) v(out)
* DC analysis
dc VIN 0 1 0.01
plot v(out) vs v(in)
.endc
.end
```

2.4 Simulation result screenshot: (Copy paste simulation result screenshots here.)

A) Transient response:



B) Voltage transfer characteristics (DC response):**2.5 Conclusion:** Comment on Switching Threshold Voltage of the Inverter (V_M)

The CMOS inverter successfully performs voltage inversion. The switching threshold voltage V_M is approximately **0.51 V**, where $V_{in} \approx V_{out}$.

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EXPERIMENT NO: 03

TITLE OF EXPERIMENT: To perform transient and DC analysis of CMOS Inverter using Cadence Virtuoso.

3.1 Aim: - To perform transient and DC analysis of CMOS Inverter using Cadence Virtuoso

3.2 Software/Hardware: - Cadence Virtuoso Schematic Editor, Spectre simulator, and gpd090 technology library.

3.3 Specifications:

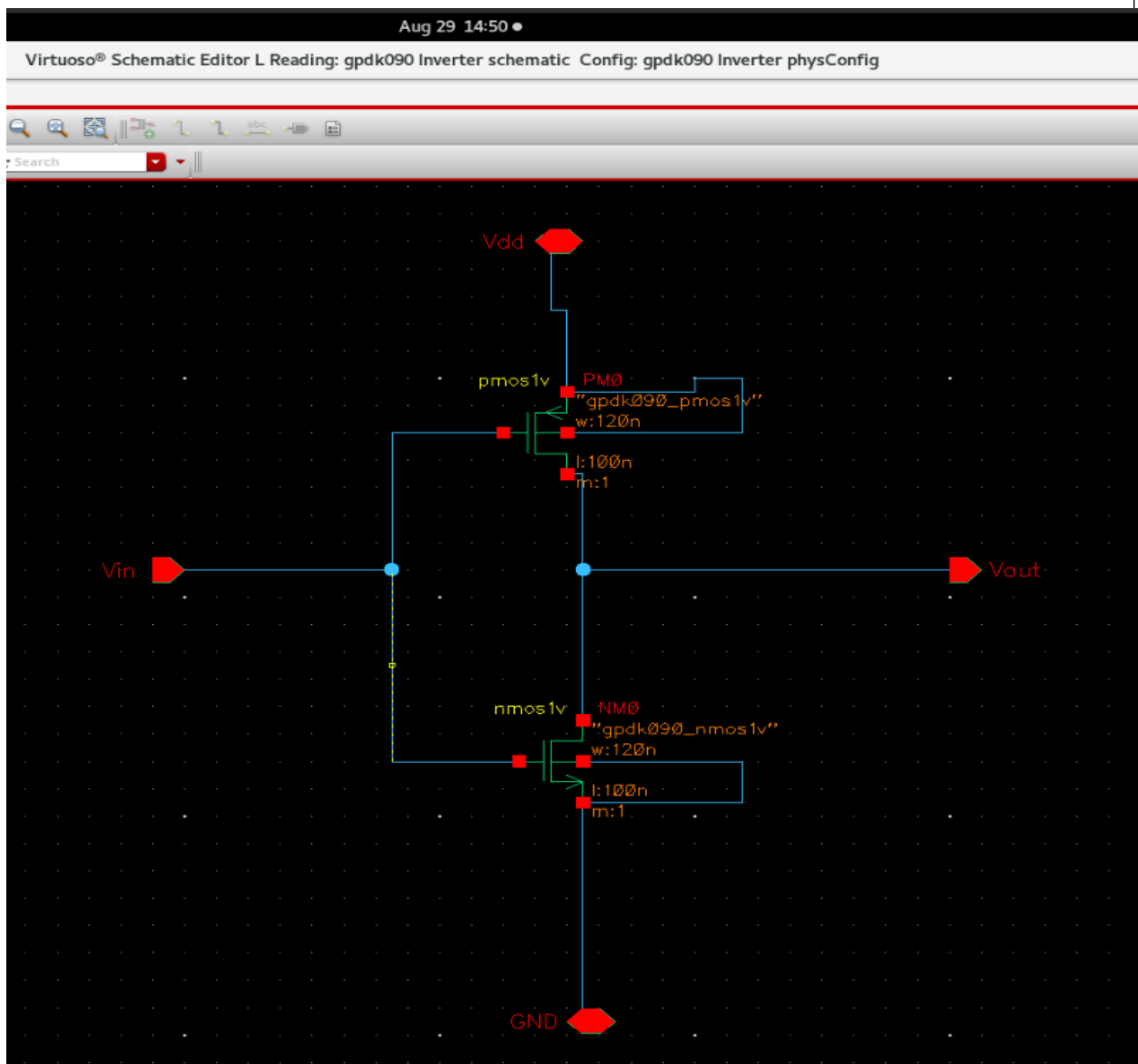
Technology: 90 nm

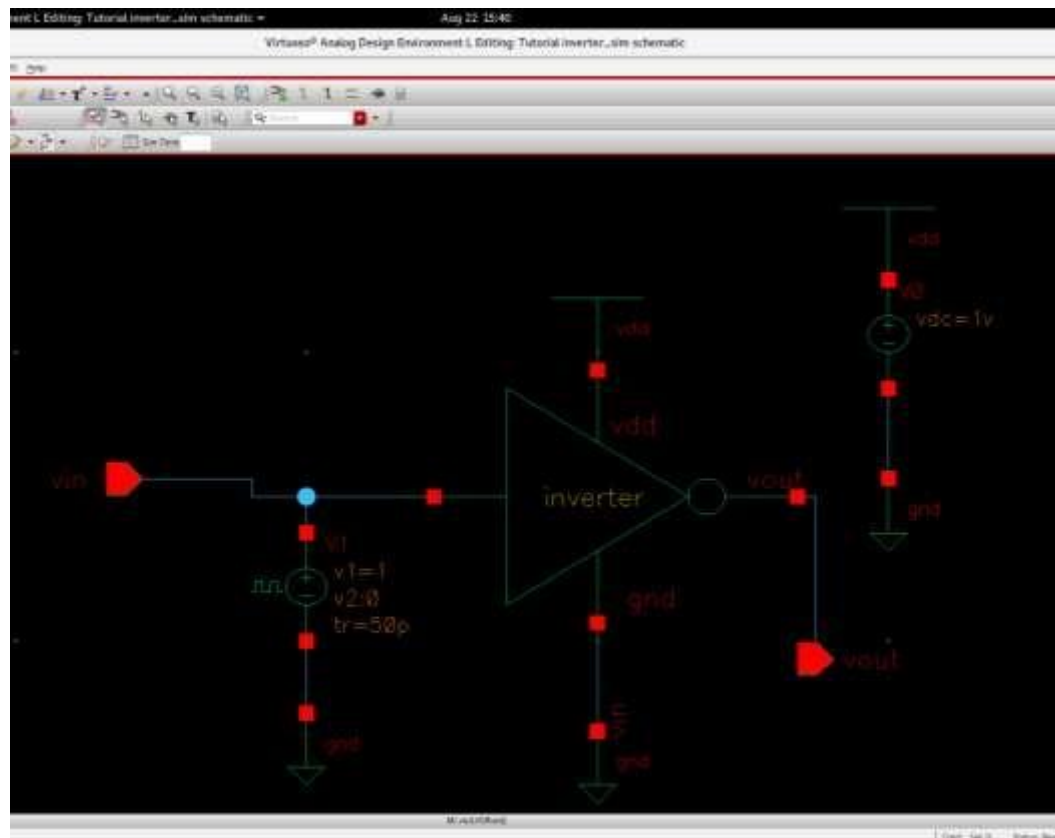
Supply voltage: 1.0V

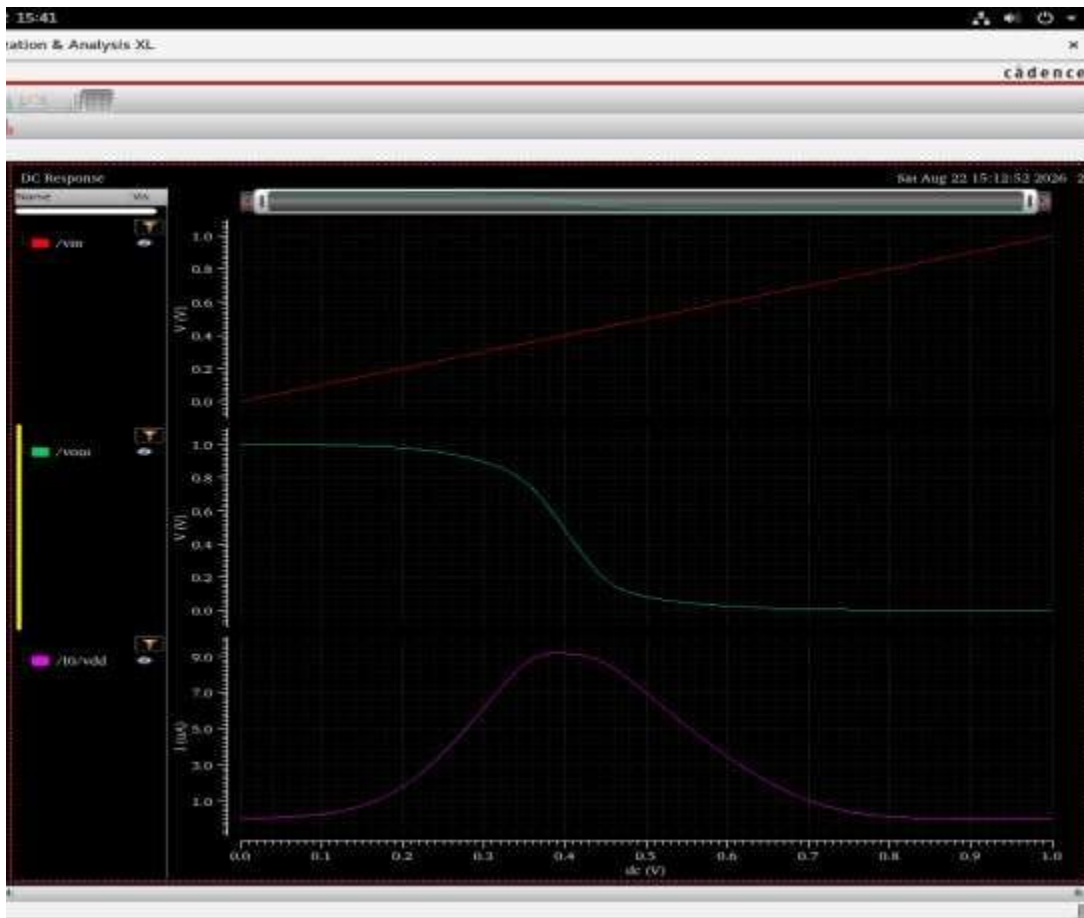
$L = 100\text{ nm}$ for both transistors, $W = 0.5\text{ }\mu\text{m}$ for NMOS and $W = 1\text{ }\mu\text{m}$ for PMOS

3.4 Schematic: (Copy paste your schematic here. Write your name and Roll. No. in comments.)

A) Circuit schematic:



B) Testbench schematic:**3.5 Simulation result screenshot: (Copy paste simulation result screenshots here.)****A) Transient response:**

B) Voltage transfer characteristics (DC response):**3.6 Observation Table:**

Specification	Value
Output high-to-low time (t_{pdf}) (in ns)	0.05 ns
Output low-to-high time (t_{pdr}) (in ns)	0.05ns
Average power dissipation (in μW)	1–2 μW

3.7 Conclusion:

The CMOS inverter was successfully designed and simulated using Cadence Virtuoso. The transient and DC responses were obtained, and the inverter showed proper voltage switching. The switching threshold voltage V_M was found to be approximately 0.5 V.

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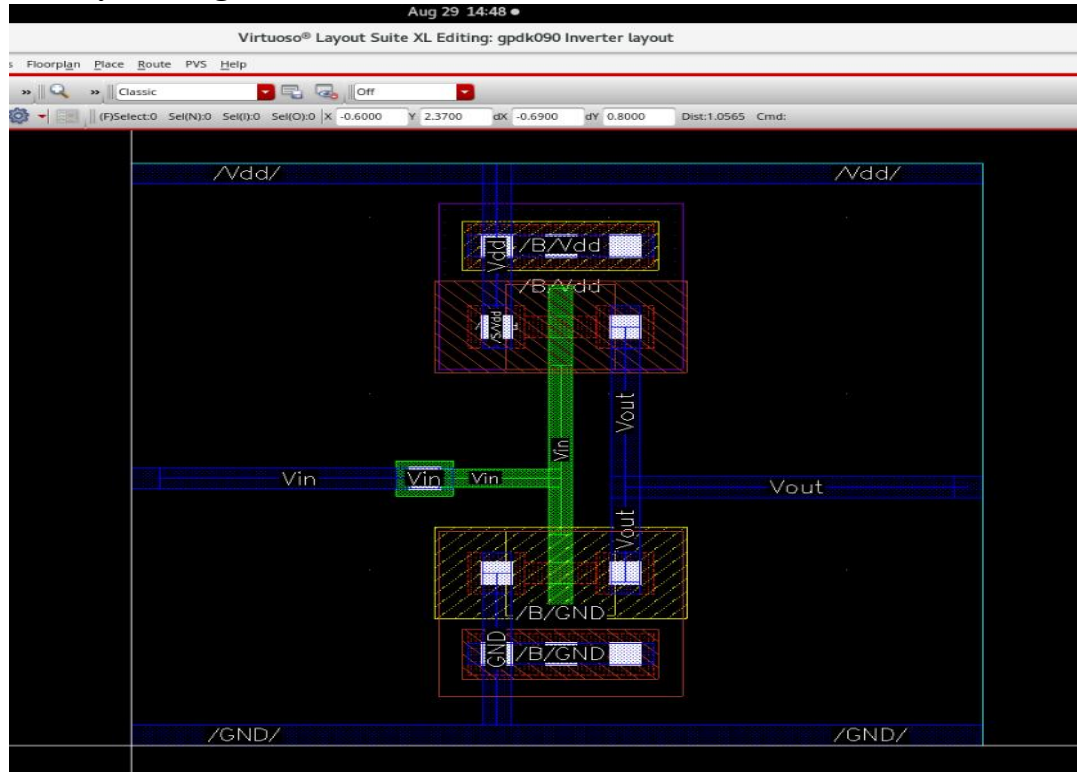
EXPERIMENT NO: 04

TITLE OF EXPERIMENT:Design of CMOS Inverter layout using Cadence Virtuoso

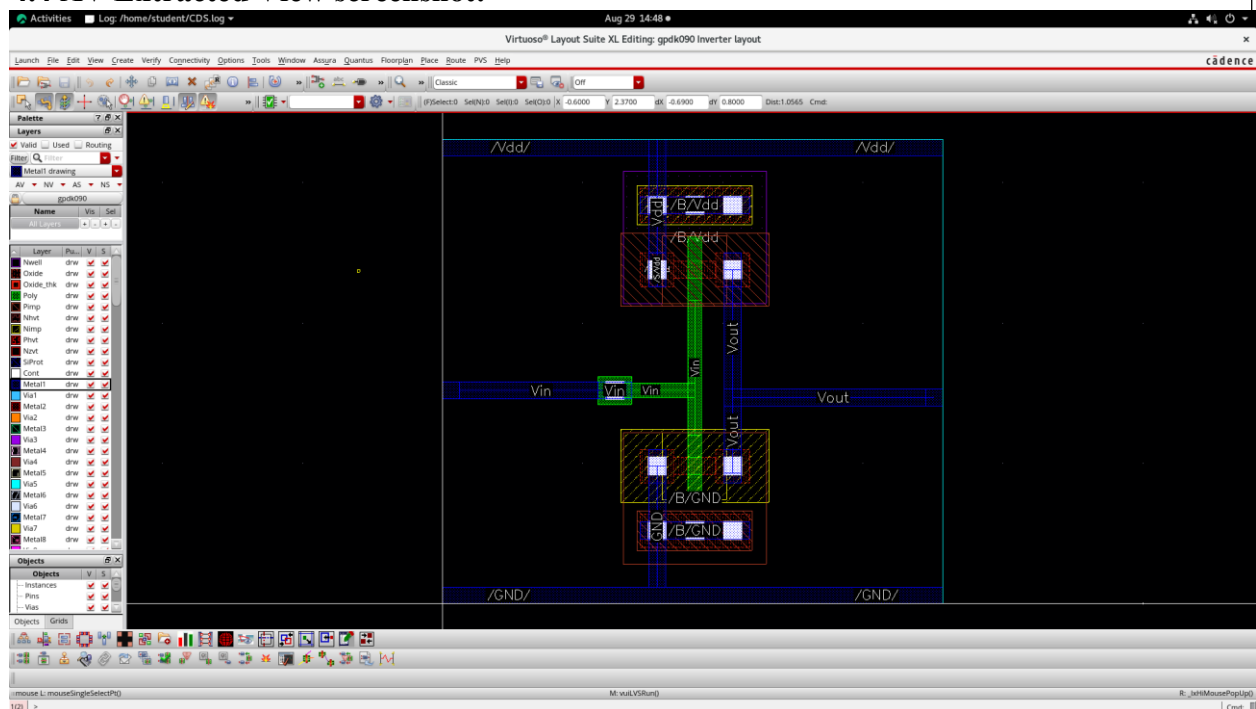
4.1 Aim: Design CMOS Inverter layout for the circuit designed in experiment number 3 using Cadence Virtuoso

4.2 Software: Cadence Virtuoso Layout Editor, Spectre simulator and gpdK090 technology library.

4.3 Layout Diagram screenshot:



4.4 AV Extracted View screenshot:



4.5 Observation:

Delay between pre-layout and post-layout simulated outputs = 0.02 ns

4.6 Conclusion:

The CMOS inverter layout was successfully designed using Cadence Virtuoso. The layout was verified using the extracted view and LVS, and the post-layout simulation confirmed proper inverter operation with a small delay due to parasitic effects.

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EXPERIMENT NO: 05

TITLE OF EXPERIMENT: To perform simulation and plot transient response of 2-input NAND Gate.

5.1 Aim: - To perform simulation and plot transient response of 2-input NAND Gate.

5.2 Software/Hardware: - Cadence Virtuoso Schematic Editor, Spectre simulator, gpdk090 technology library.

5.3 Specifications:

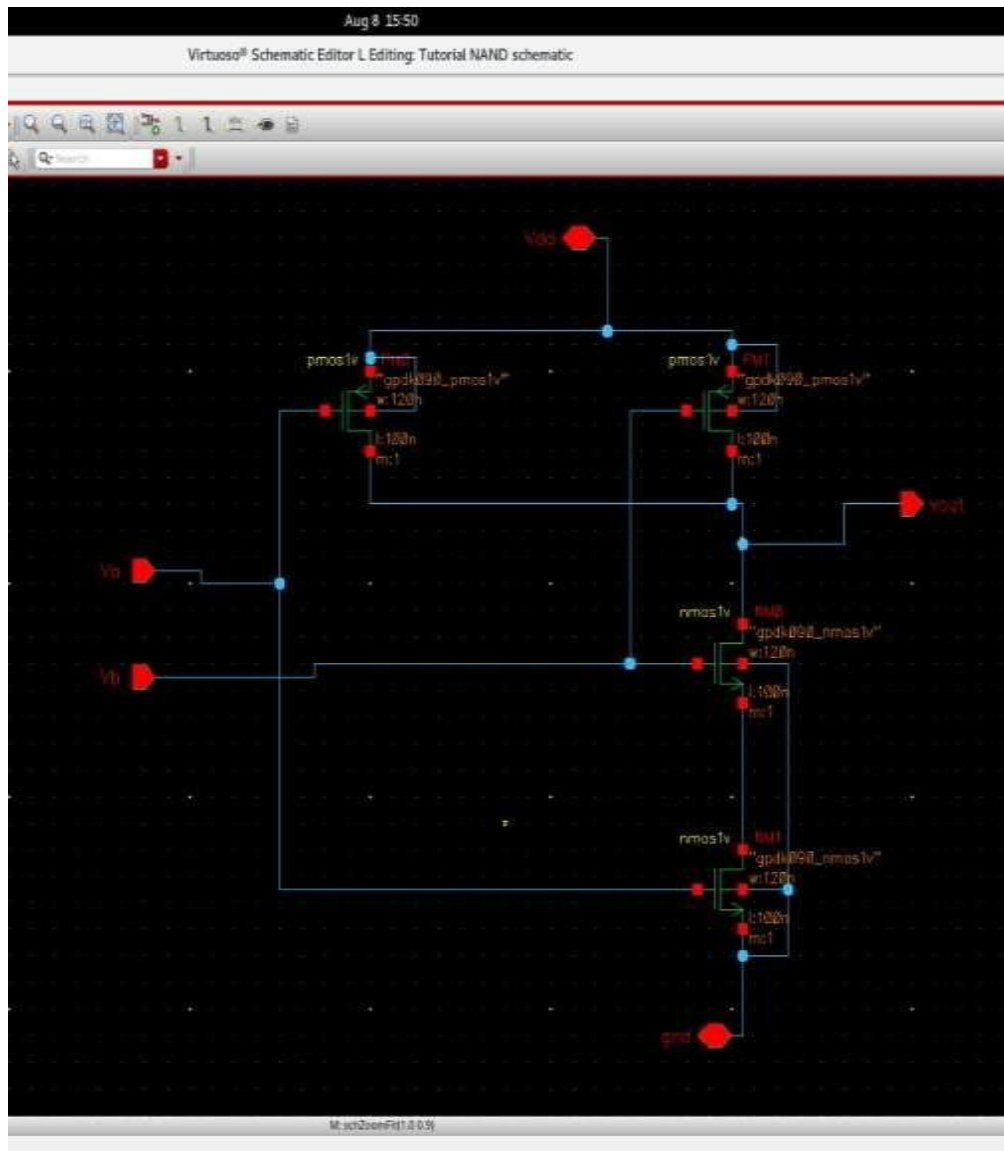
Technology: 90 nm

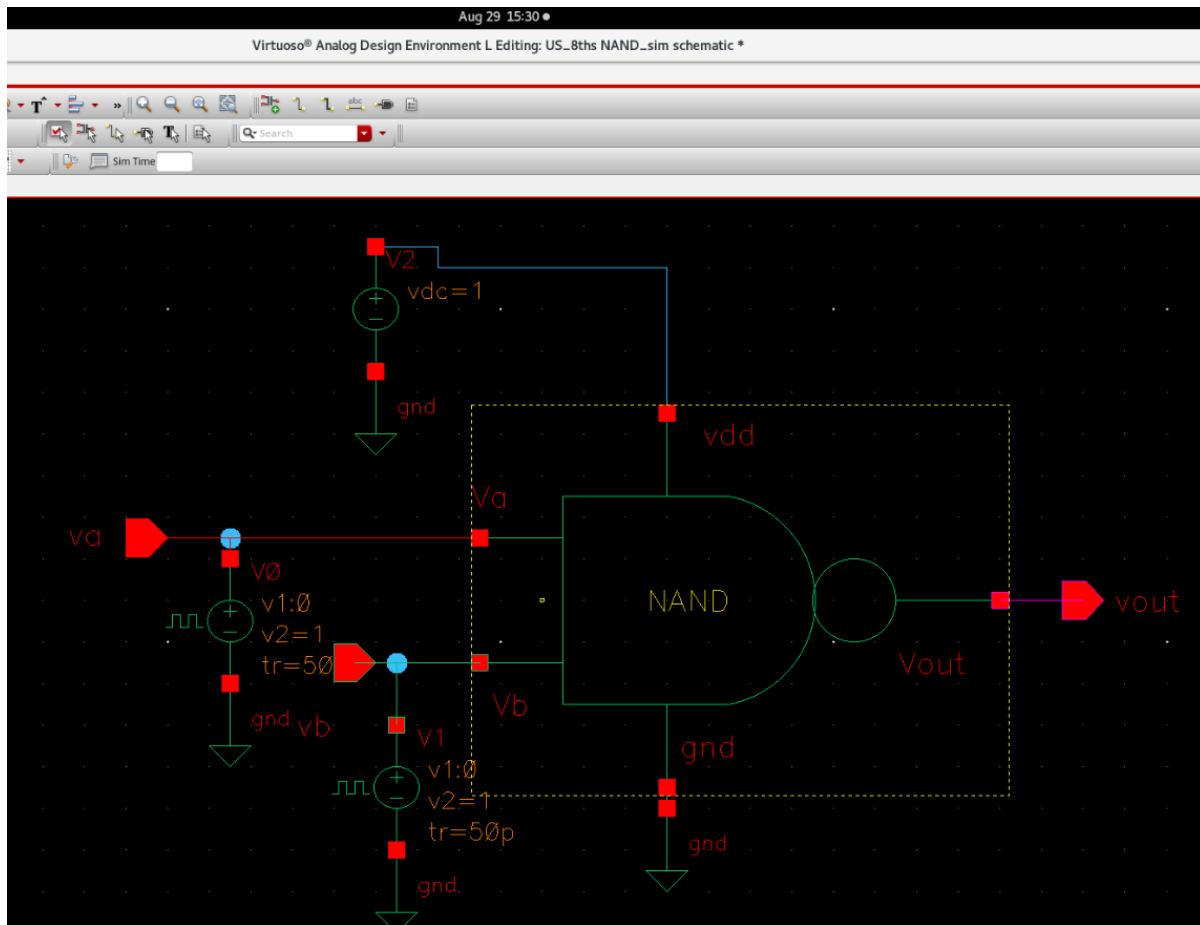
Supply voltage: 1.0V

L = 100 nm for all transistors, W = 480 nm for all transistors

5.4 Schematic: (Copy paste your schematic here. Write your name and Roll. No. in comments.)

A) NAND gate schematic:



B) NAND gate testbench:

5.5 Simulation result screenshot: (Copy paste simulation result screenshots here.)

Transient response of NAND gate:

5.6 Observation Table for NAND gate:

Specification	Value
Input 1 to Output high-to-low time (t_{pdf}) (in ns)	0.1 ns
Input 1 to Output low-to-high time (t_{pdr}) (in ns)	0.1 ns
Input 2 to Output high-to-low time (t_{pdf}) (in ns)	0.1 ns
Input 2 to Output low-to-high time (t_{pdr}) (in ns)	0.1 ns
Average power dissipation (in μ W)	10 uW

5.7 Conclusion:

The 2-input CMOS NAND gate was successfully designed and simulated using Cadence Virtuoso. The transient response verified the NAND logic operation, where the output becomes LOW only when both inputs are HIGH. The gate showed proper switching with small propagation delay and power dissipation.

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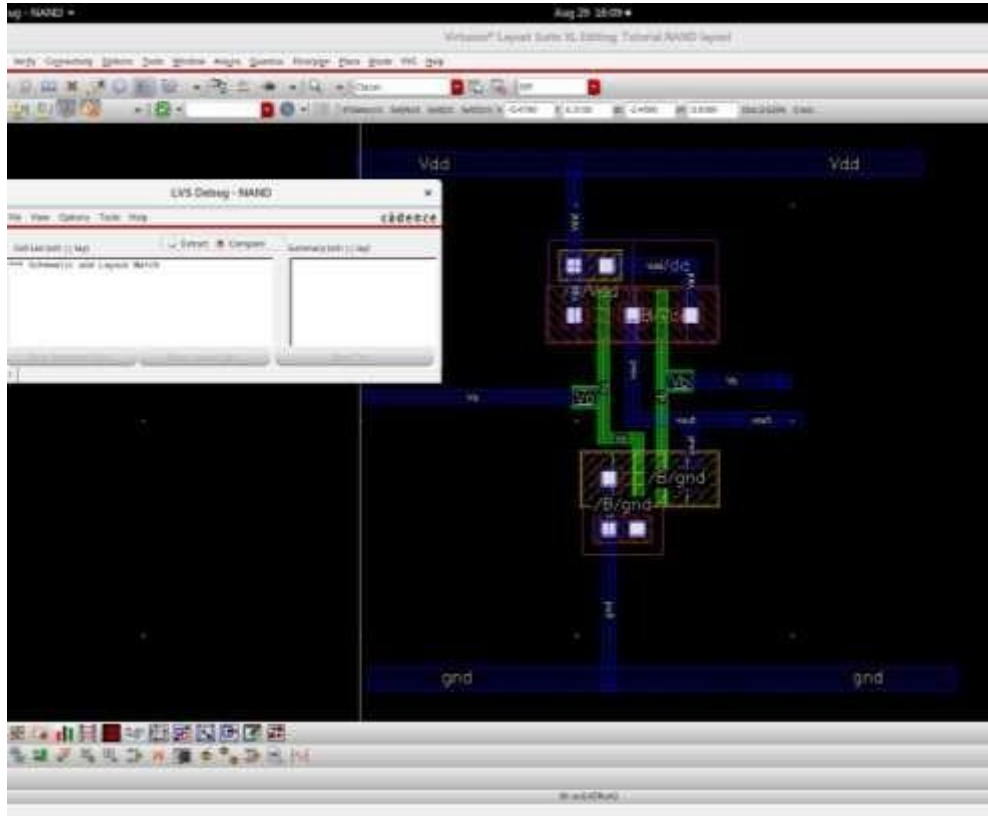
Division: G

EXPERIMENT NO: 06
TITLE OF EXPERIMENT: Design of CMOS 2-input NAND gate layout.

6.1 Aim: Design of CMOS 2-input NAND gate layout.

6.2 Software: Cadence Virtuoso Layout Editor, Spectre simulator, and gpd90 technology library

6.3 Layout Diagrams of NAND gate:



6.4 AV Extracted View of NAND gate:



6.5 Conclusion:

The 2-input CMOS NAND gate layout was successfully designed and verified using Cadence Virtuoso. The layout showed correct connections for VDD, GND, inputs, and output, and the LVS verification passed successfully, confirming that the layout matches the schematic.